

# Independent calculations of fault valving: interim results

Reproducibility study of Zhu, Allison, Dunham, and Yang (2020)

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**This is an interim scientific report. The full study remains incomplete.** At this snapshot, 11 of 12 configured cases have completed, and 49 of 49 panel packages are available. Of these, 2 are independently reproduced constitutive-law panels and 47 are partial comparisons, including one newly drawn schematic. Pending runs are `baseline_fine`. Their configured durations remain unchanged.

## 1 Scope and independence

Fault slip increases permeability, while healing between earthquakes restricts drainage. The resulting pore-pressure changes alter frictional strength. We independently implemented the coupled earthquake-cycle and fluid-flow model of Zhu et al. [1] in C++. The scientific inputs are the published article, its supplement, and comparable written model documentation. The earlier project attempt used authors' archived outputs; that attempt and all its simulation inputs are excluded here. No authors' code, simulation output, source dataset, digitized output curve, or published image supplies a numerical result or figure in this report. The schematic is newly drawn and is not a reproduced simulation.

Every displayed numerical panel comes from independent arrays in `data/panels`. Its JSON sidecar records parameters, source hashes, selection window, and provenance. A constitutive-law check establishes a narrower result than recovery of an earthquake sequence. Displayed cycle calculations retain the status in each inventory row because timing, event patterns, assumed inputs, or spatial convergence remain unresolved. Published cycle durations below are approximate visual readings used only as comparison targets, never computational inputs.

## 2 Physical model and declared assumptions

The elastic domain occupies  $0 \leq y \leq L_y$  and  $0 \leq z \leq L_z$ , with depth  $z$  positive downward. Both dimensions are 500 km. Antiplane displacement  $u$  is half the fault slip on  $y = 0$ ; the remote side is displaced at half the plate rate. The top and bottom are traction free. The shear modulus is 32.4 GPa,  $V_p = 10^{-9} \text{ m s}^{-1}$ , and the radiation-damping coefficient is  $\eta_{\text{rad}} = 4.68 \times 10^6 \text{ Pa s m}^{-1}$ .

The regularized friction law and aging law are

$$f(V, \psi) = a \operatorname{asinh} \left[ \frac{V}{2V_0} \exp \left( \frac{\psi}{a} \right) \right], \quad (1)$$

$$\dot{\psi} = \frac{b}{d_c} \left[ V_0 \exp \left( \frac{f_0 - \psi}{b} \right) - |V| \right], \quad (2)$$

$$\tau_{\text{qs}} - \eta_{\text{rad}} V = (\sigma - p) f(V, \psi). \quad (3)$$

Here  $V_0 = 10^{-6} \text{ m s}^{-1}$ ,  $f_0 = 0.6$ , and  $d_c = 2 \text{ mm}$ . Friction is signed; slip speed is used in state and permeability evolution if reverse slip occurs. This extends the paper's positive-sliding

convention and is documented explicitly. Radiation damping limits rapid slip within the quasi-dynamic approximation; it does not reproduce propagating elastic waves.

Fault-zone fluid storage, Darcy flux, and permeability obey

$$n\beta \partial_t p = \partial_z q, \quad q = \frac{k}{\eta} (\partial_z p - \rho g), \quad (4)$$

$$k = k_{\min} + (k_* - k_{\min}) \exp[-(\sigma - p)/\sigma_*], \quad (5)$$

$$\dot{k}_* = \frac{|V|}{L} (k_{\max} - k_*) - \frac{k_* - k_{\min}}{T}. \quad (6)$$

Positive  $q$  denotes upward flow. The surface pressure is zero and the bottom influx is fixed. We use  $n\beta = 10^{-11} \text{ Pa}^{-1}$ ,  $\eta = 10^{-4} \text{ Pa s}$ ,  $\rho = 1000 \text{ kg m}^{-3}$ ,  $g = 9.8 \text{ m s}^{-2}$ ,  $\sigma_* = 30 \text{ MPa}$ ,  $L = 1 \text{ m}$ ,  $k_{\min} = 10^{-19} \text{ m}^2$ , and  $k_{\max} = 10^{-15} \text{ m}^2$ . The healing times are  $10^7$ ,  $10^8$ ,  $10^9$ , and  $10^{10} \text{ s}$ . Bottom influx is  $3 \times 10^{-9} \text{ m s}^{-1}$  except for  $T = 10^7 \text{ s}$ , where it is  $3.3 \times 10^{-10} \text{ m s}^{-1}$ . Porosity changes, thermal pressurization, and bulk poroelastic stress changes are omitted as in the published model.

Several inputs needed for an executable model are not fully specified in the article or supplement. We assume  $\sigma(z) = (22 \text{ MPa km}^{-1})z$ , consistent with the displayed normal-stress line in the original Fig. 1d. We prescribe  $a = 0.01$  at the surface, increasing linearly to 0.03 at 15 km, then increasing at  $0.003 \text{ km}^{-1}$ . We set  $a - b = -0.01$  above 13.5 km and increase it linearly below that depth, crossing zero at 17 km. These profiles continue to the bottom boundary. They approximate a parameter diagram, not a digitized simulation result, and their exact agreement with the authors' inputs cannot be established from the written record.

Initially  $k_*$  has its steady-sliding value at  $V_p$ , and pressure is the discrete steady solution with flux  $q_0$  at every face. State is initialized at its steady-sliding value at  $V_p$ . Initial velocity varies from  $10^{-4}V_p$  above 17 km to  $V_p$  below, through a hyperbolic tangent of width 1.5 km. Elastic prestress balances that initial velocity. A Gaussian reduction of  $10^{-4}$  in  $\psi$ , centered at 12 km with width 1 km, perturbs the fault. Earthquake times, rupture footprints, and subsequent slip are not prescribed. These initial conditions are independent choices; no published restart is used.

An exploratory all-depth steady-sliding initialization was also tested. Shallow elastic stiffness imposed a severe explicit time-step restriction. The nearly locked shallow start is therefore used for the production runs; its influence is assessed separately from numerical resolution. Full parameter files and all choices are included in the repository.

### 3 Independent numerical method

The elastic boundary-value problem can be reduced analytically to a fault stiffness operator. A cosine mode with  $\kappa_j = j\pi/L_z$  has stiffness

$$K_0 = \frac{\mu}{2L_y}, \quad K_j = \frac{\mu\kappa_j}{2} \coth(\kappa_j L_y), \quad j > 0. \quad (7)$$

The factor one-half converts fault slip to one-sided displacement. These eigenvalues retain the finite side and bottom boundaries. FFTW cosine transforms apply this operator on cell centers, with  $\tau_{\text{qs}} = \tau_{\text{initial}} - Kd$  and  $d$  equal to accumulated slip minus  $V_p t$ .

Physical fault nodes form a graded subset of this auxiliary grid. If  $P$  linearly interpolates physical slip to the cosine grid, the physical operator is  $W^{-1}P^T h K P$ , where  $W$  contains control-volume widths and  $h$  is the auxiliary spacing. This weighted Galerkin projection preserves constant traction, elastic reciprocity, and positive elastic energy. It retains the fine grid through the rupture region while reducing the number of far-depth unknowns.

Pressure is discretized by conservative finite volumes, using excess pressure  $p - \rho g z$  and harmonic face permeability. Gradients use actual node distances; surface pressure is imposed at the surface and

bottom flux enters storage directly. This scheme differs from the paper’s fourth-order summation-by-parts method. The production cosine grid has 131,072 points, giving a minimum fault spacing of 3.815 m. Spacing is finest through the rupture depths and down to 30 km; it increases at greater depths toward a 1 km target. The final interval can reach about 1.5 times that target to avoid an undersized boundary interval. The surface spacing is about eight times the minimum and decreases to the minimum over the upper 2 km. Both the minimum and maximum spacings halve or double in refinement cases. The number of physical nodes and the resolved frictional length scales are recorded in the completed-case audit files. No stress or velocity floor replaces spatial resolution.

We use the fourth-order additive Runge–Kutta method ARK4(3)6L[2]SA [3], treating fluid flow implicitly and slip, state, and  $k_*$  explicitly. The nonlinear pressure solve iterates permeability and a tridiagonal solve to an absolute pressure-update tolerance of 0.001 Pa. An embedded third-order estimate controls errors in slip, state, permeability, and pressure. The fourth-order solution is accepted. Production tolerance is  $10^{-5}$  in the scaled maximum norm; the time-accuracy case uses  $2.5 \times 10^{-6}$ . A controlled interseismic benchmark showed that this tighter tolerance reduced accumulated errors without substantially increasing the cost of the graded calculation. The maximum step is  $10^6$  s. Detailed coefficients, scales, boundary formulas, and data formats are given in `documentation/numerics.md`.

## 4 Completed runs and quantitative comparisons

Every completed case passed an audit of all accepted history rows and saved field profiles, including samples outside the figure windows. The audit checks finite values, positive effective stress, permeability bounds, accepted error estimates, consecutive steps, field/history consistency, and source and parameter hashes. Fixed-pressure cases also retain pressure throughout their saved profiles. These checks establish internal consistency; they do not establish sequence convergence or agreement with the paper. The audit files are `data/<case>/output_verification.json`.

| Case               | Duration (yr) | Accepted steps | Profiles | Large events |
|--------------------|---------------|----------------|----------|--------------|
| reference          | 400           | 164586         | 14041    | 5            |
| baseline           | 200           | 624522         | 16510    | 7            |
| short              | 200           | 272347         | 8873     | 4            |
| long               | 350           | 306854         | 17162    | 6            |
| long_reference     | 350           | 209873         | 12897    | 4            |
| verylong           | 500           | 216379         | 18414    | 3            |
| verylong_reference | 500           | 334274         | 19194    | 4            |
| baseline_coarse    | 200           | 438564         | 14838    | 7            |
| baseline_tight     | 200           | 736538         | 16455    | 6            |
| initialization     | 100           | 282753         | 7219     | 4            |
| normal_stress      | 100           | 287400         | 7227     | 4            |

A seismic interval is a whole-fault maximum speed excursion above  $10^{-3}$  m s $^{-1}$ . A resolved rupture accumulates at least 1 cm of slip in a connected depth interval. A large event has a connected footprint reaching above 2 km and spanning more than 10 km. Slip is measured between the saved first-above and first-below threshold profiles. Incomplete events cannot define cycles. Velocity maps use the last complete large-event cycle; cumulative-slip profiles use the last two complete cycles. All available recurrence intervals are retained in each analysis file. Neither event selection nor time windows are fitted to the published images.

| Case               | Last cycle (yr) | All-cycle median (yr) | Paper (yr) | Difference (%) |
|--------------------|-----------------|-----------------------|------------|----------------|
| reference          | 64              | 66                    | 50         | 27.9           |
| baseline           | 31.4            | 27.1                  | 32         | -1.76          |
| long_reference     | 88.4            | 88.4                  | 160        | -44.8          |
| long               | 70.5            | 63.6                  | 65         | 8.4            |
| verylong_reference | 95.1            | 95.2                  | 160        | -40.6          |
| verylong           | 174             | 179                   | 150        | 16.3           |

The reference and baseline targets are main Figs. 2a and 2c; the long-healing targets are supplementary Figs. 3a,c and 5a,c. The differences describe the selected final cycles and do not imply stationary recurrence. In particular, the two long-healing fixed-pressure references differ substantially from their approximately 160-year targets. Uncertain friction profiles and initialization, finite simulated duration, and unestablished spatial convergence prevent assigning these discrepancies to one cause. No high-healing-time mesh study has been completed.

| Case     | $T$ (s) | $\Delta(\sigma - p)$ at 10 km (MPa) | Leading front<br>(m/yr) | Deepest front<br>(m/yr) |
|----------|---------|-------------------------------------|-------------------------|-------------------------|
| short    | 1e+07   | 4.57                                | 1.84e+03                | 1.53e+03                |
| baseline | 1e+08   | 28.1                                | 647                     | 967                     |
| long     | 1e+09   | 16.5                                | 187                     | 242                     |
| verylong | 1e+10   | 6.82                                | 168                     | 168                     |

The stress range uses the nearest saved depth, about 10.004 km. Front rates are medians of all retained upward segments of the  $|V| = V_p$  contour, with  $R^2 \geq 0.8$ , at 13–20 km depth. The leading and deepest branches are contour extrema, not tracked identities of individual pulses. Their definitions need not coincide with the paper’s reported migration speeds of 2500, 380, 120, and 30 m/yr for the short, baseline, long, and very-long healing cases. The discrepancies are retained rather than selecting the branch closest to the published rate.

The baseline recurrence intervals range from 25.4 to 54.6 yr; the close agreement of its final interval with the displayed 32-year example does not establish a stationary cycle. At the saved depth nearest 10 km, effective stress spans 19.1–47.2 MPa, a variation of 28.1 MPa compared with the article’s 10–20 MPa scale. The maximum occurs 19.4 yr after cycle opening. This maximum is a diagnostic, not an identification of the paper’s phase boundary. Maximum upward flux there is 3.67e-08 m/s, compared with the article’s order  $10^{-7}$  m/s upper scale.

The first two resolved partial ruptures are retained in chronological order: 0.132 yr after opening, with a connected footprint at 0.0134–5.74 km; 18 yr after opening, with a connected footprint at 12.1–17.1 km. The extra early shallow rupture differs from the published sequence. The final two-year window contains 5 smaller ruptures intersecting 2–10 km. This recovers swarm-like behavior, while its timing and depth range differ. The closing large event first crosses the speed threshold at 6.1 km, and its connected footprint spans 0.0134–11.5 km. The paper gives a 12 km nucleation depth for the preceding 8–19 km rupture; it does not specify a numerical nucleation depth for the final surface-breaking event in this description. Our threshold definition does not establish identical published event boundaries.

#### 4.1 Depth histories, alternate cycles, and event labels

Supplementary Fig. 1 retains 168789 accepted samples at the nearest physical nodes to 5, 10, 15, and 20 km. Minimum slip speeds at the first two depths are 9.61e-23 and 1.5e-20 m/s. These calculated minima and the timing of the transients provide additional comparisons beyond the final recurrence interval. Supplementary Figs. 2a and 2b use fixed earlier cycles of 26.8 and 54.6 yr, containing 8 and 23 smaller resolved ruptures, respectively. Their event catalog and complete windows are

retained even when their behavior differs from the alternate sequences described in the published supplement. The large-event label requires a connected footprint reaching above 2 km and spanning more than 10 km. A postprocessing sensitivity check changes that span threshold to 9, 10, and 11 km and obtains 8, 7, and 7 large events. The simulation and panel windows remain unchanged; all primary comparisons retain the declared 10 km criterion. The surface-reaching event at year 118 has a 0.0134–9.49 km connected footprint; the 10 km criterion excludes it. Recurrence statistics therefore depend on the event definition as well as the numerical trajectory.

The short-healing case produces aseismic pulses. The following medians compare with the article’s approximate one-year duration, few-year recurrence, and few-centimeter slip scales. Pulse widths depend on our declared threshold,  $|V| \geq 1.1V_p$ . Each counted episode lasts at least 0.01 yr, lies within the selected window, and remains below the seismic threshold at every accepted whole-fault history sample. All qualifying episodes are retained in the case analysis. The depth dependence limits any claim of agreement.

| Depth (km) | Episodes | Duration (yr) | Recurrence (yr) | Slip (cm) |
|------------|----------|---------------|-----------------|-----------|
| 15         | 14       | 0.302         | 3.04            | 4.72      |
| 18         | 11       | 0.611         | 3.17            | 2.46      |
| 20         | 2        | 0.894         | 8.64            | 3.19      |

## 4.2 Temporal-tolerance comparison

The time-tolerance comparison also includes every resolved rupture over the first 100 years. The baseline and tighter-tolerance cases contain 20 and 20 resolved ruptures, respectively, from 21 and 20 complete threshold excursions. The declared large-event criterion identifies 4 and 4 events, respectively. Across the 20 chronological pairs, the maximum absolute differences are 83.2 min in onset, 1.47% in duration, 1.1% in peak speed, and 0.0062% in maximum slip. All pairs have identical connected 1 cm rupture footprints on the saved depth grid and identical large-event classifications. These measurements test time tolerance on the baseline mesh. Footprint comparisons are limited by the saved depth spacing, about 30 m in the scientific runs. They do not establish spatial convergence or recovery of the published swarm sequence.

The baseline and tighter-tolerance runs both extend beyond the common validation window. We therefore repeat the comparison over their full shared duration of 200 years, so agreement confined to the early cycles cannot substitute for assessment of later events. The time-tolerance comparison also includes every resolved rupture over the first 200 years. The baseline and tighter-tolerance cases contain 60 and 58 resolved ruptures, respectively, from 61 and 58 complete threshold excursions. The declared large-event criterion identifies 7 and 6 events, respectively. Across the 58 chronological pairs, the maximum absolute differences are  $1.63\text{e}+07$  min in onset, 718% in duration, 743% in peak speed, and 852% in maximum slip. Counts, saved rupture footprints, or large-event classifications differ between the catalogs; chronological pairing alone does not establish physical correspondence. The reported maxima compare ordinal catalog positions and cannot be interpreted as timing or amplitude errors between physically matched events. These measurements test time tolerance on the baseline mesh. Footprint comparisons are limited by the saved depth spacing, about 30 m in the scientific runs. They do not establish spatial convergence or recovery of the published swarm sequence.

These long-duration catalog differences preclude claiming convergence of individual late events from this tolerance test. Early agreement does not certify the later figure cycles. Large-event counts also depend on the declared span cutoff. With a 9 km cutoff, the baseline and tighter-tolerance catalogs contain 8 and 8 large events, compared with 7 and 6 at 10 km. All primary results retain 10 km; this relabeling does not remove differences in the resolved small-event sequences. The fine-mesh and input-sensitivity runs do not cover this longer window. This additional temporal comparison therefore does not establish spatial convergence of the later figure cycles.

The final model assessment requires the baseline, mesh-refined case, and tighter time-tolerance case together. This interim report does not treat completed coarse-grid, normal-stress, or initialization runs as a completed sensitivity comparison. Software fixtures and exploratory short calculations are excluded from all scientific figures here.

## 5 Panel inventory and provenance

Each row specifies the original label, quantity, documentation, independent-data command, plotting command, status, and discrepancy. Data paths follow `data/panels/<label>.npz`, with parameters and provenance in the adjacent JSON file. Commands are intended for a clean checkout or completed cases; do not start a duplicate simulation in a directory containing a live run. Supplementary Fig. 2 visibly labels rows a and b; its caption also mentions c and d, which are not separately visible. The inventory therefore includes the two displayed rows and records that inconsistency.

| Panel | Quantity                                           | Documentation                                                    | Data command                                     | Plot command                             | Status                   | Discrepancy                                                                                                                                                                                                              |
|-------|----------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------|------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F1a   | Geometry schematic and assumed friction profiles   | Main Fig. 1a; Eqs. (7)-(9); Table 1; $a(z)$ , $b(z)$ are assumed | <code>python3 scripts/panel.py F1a --data</code> | <code>python3 scripts/plot.py F1a</code> | partial                  | New schematic, not a simulation reproduction; friction profiles are declared approximations.                                                                                                                             |
| F1b   | Permeability versus effective normal stress        | Main Eq. (3), Fig. 1b; Table 1                                   | <code>python3 scripts/panel.py F1b --data</code> | <code>python3 scripts/plot.py F1b</code> | independently reproduced | Analytic laws agree with numerical limiting-case verification.                                                                                                                                                           |
| F1c   | Healing and slip-enhancement limiting solutions    | Main Eq. (4), Fig. 1c; Table 1                                   | <code>python3 scripts/panel.py F1c --data</code> | <code>python3 scripts/plot.py F1c</code> | independently reproduced | Analytic laws agree with numerical limiting-case verification.                                                                                                                                                           |
| F1d   | Steady pressure, stress, and permeability profiles | Main Eqs. (2)-(4), Fig. 1d; assumed normal-stress gradient       | <code>python3 scripts/panel.py F1d --data</code> | <code>python3 scripts/plot.py F1d</code> | partial                  | Normal stress gradient is assumed; steady Darcy balance is independently verified.                                                                                                                                       |
| F2a   | Slip velocity in time and step coordinates         | Main Fig. 2; main Eqs. (1)-(9), Table 1                          | <code>python3 scripts/panel.py F2a --data</code> | <code>python3 scripts/plot.py F2a</code> | partial                  | Last complete interval 64 yr versus approximately 50 yr in Main Fig. 2a (+28%). All-interval median 66 yr. Initial state and friction profiles remain uncertain.                                                         |
| F2b   | Cumulative slip profiles                           | Main Fig. 2; main Eqs. (1)-(9), Table 1                          | <code>python3 scripts/panel.py F2b --data</code> | <code>python3 scripts/plot.py F2b</code> | partial                  | Last complete interval 64 yr versus approximately 50 yr in Main Fig. 2a (+28%). All-interval median 66 yr. Initial state and friction profiles remain uncertain.                                                         |
| F2c   | Slip velocity in time and step coordinates         | Main Fig. 2; main Eqs. (1)-(9), Table 1                          | <code>python3 scripts/panel.py F2c --data</code> | <code>python3 scripts/plot.py F2c</code> | partial                  | Last complete interval 31.4 yr versus approximately 32 yr in Main Fig. 2c (-2%). All-interval median 27.1 yr. Initial state and friction profiles remain uncertain. First partial footprints: 0.01–5.74, 12.13–17.15 km. |
| F2d   | Cumulative slip profiles                           | Main Fig. 2; main Eqs. (1)-(9), Table 1                          | <code>python3 scripts/panel.py F2d --data</code> | <code>python3 scripts/plot.py F2d</code> | partial                  | Last complete interval 31.4 yr versus approximately 32 yr in Main Fig. 2c (-2%). All-interval median 27.1 yr. Initial state and friction profiles remain uncertain. First partial footprints: 0.01–5.74, 12.13–17.15 km. |
| F3a   | Effective normal stress history                    | Main Fig. 3; main Eqs. (1)-(4), Table 1                          | <code>python3 scripts/panel.py F3a --data</code> | <code>python3 scripts/plot.py F3a</code> | partial                  | At 10 km, stress varies by 28.1 MPa; article gives an order 10–20 MPa scale. Phase timing differs.                                                                                                                       |
| F3b   | Effective normal stress depth profiles             | Main Fig. 3; main Eqs. (1)-(4), Table 1                          | <code>python3 scripts/panel.py F3b --data</code> | <code>python3 scripts/plot.py F3b</code> | partial                  | At 10 km, stress varies by 28.1 MPa; article gives an order 10–20 MPa scale. Phase timing differs.                                                                                                                       |
| F3c   | Permeability history                               | Main Fig. 3; main Eqs. (1)-(4), Table 1                          | <code>python3 scripts/panel.py F3c --data</code> | <code>python3 scripts/plot.py F3c</code> | partial                  | At 10 km, $k$ spans 4.32e-19–3.58e-16 square metres; exact cycle phases are not recovered.                                                                                                                               |
| F3d   | Permeability depth profiles                        | Main Fig. 3; main Eqs. (1)-(4), Table 1                          | <code>python3 scripts/panel.py F3d --data</code> | <code>python3 scripts/plot.py F3d</code> | partial                  | At 10 km, $k$ spans 4.32e-19–3.58e-16 square metres; exact cycle phases are not recovered.                                                                                                                               |
| F3e   | Upward fluid flux history                          | Main Fig. 3; main Eqs. (1)-(4), Table 1                          | <code>python3 scripts/panel.py F3e --data</code> | <code>python3 scripts/plot.py F3e</code> | partial                  | At 10 km, maximum flux is 3.67e-08 m/s; the article gives an order 1e-7 m/s upper scale. Phase timing differs.                                                                                                           |

| Panel | Quantity                                  | Documentation                                                             | Data command                                     | Plot command                             | Status  | Discrepancy                                                                                                                                                                                                                                                          |
|-------|-------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------|------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F3f   | Upward fluid flux depth profiles          | Main Fig. 3; main Eqs. (1)-(4), Table 1                                   | <code>python3 scripts/panel.py F3f --data</code> | <code>python3 scripts/plot.py F3f</code> | partial | At 10 km, maximum flux is 3.67e-08 m/s; the article gives an order 1e-7 m/s upper scale. Phase timing differs.                                                                                                                                                       |
| F4a   | Slip velocity versus time                 | Main Fig. 4; Eqs. (1)-(9); Table 1                                        | <code>python3 scripts/panel.py F4a --data</code> | <code>python3 scripts/plot.py F4a</code> | partial | 5 small ruptures intersect 2–10 km in the fixed two-year window; exact swarm sequence differs.                                                                                                                                                                       |
| F4b   | Slip velocity versus accepted step        | Main Fig. 4; Eqs. (1)-(9); Table 1                                        | <code>python3 scripts/panel.py F4b --data</code> | <code>python3 scripts/plot.py F4b</code> | partial | 5 small ruptures intersect 2–10 km in the fixed two-year window; exact swarm sequence differs.                                                                                                                                                                       |
| F4c   | Effective normal stress                   | Main Fig. 4; Eqs. (1)-(9); Table 1                                        | <code>python3 scripts/panel.py F4c --data</code> | <code>python3 scripts/plot.py F4c</code> | partial | 5 small ruptures intersect 2–10 km in the fixed two-year window; exact swarm sequence differs.                                                                                                                                                                       |
| F4d   | Permeability                              | Main Fig. 4; Eqs. (1)-(9); Table 1                                        | <code>python3 scripts/panel.py F4d --data</code> | <code>python3 scripts/plot.py F4d</code> | partial | 5 small ruptures intersect 2–10 km in the fixed two-year window; exact swarm sequence differs.                                                                                                                                                                       |
| F5a   | Slip velocity versus time                 | Main Fig. 5; Eqs. (1)-(9); Table 1; $T=1e7$ s, $q_0=3.3e-10$ m/s          | <code>python3 scripts/panel.py F5a --data</code> | <code>python3 scripts/plot.py F5a</code> | partial | Complete aseismic episodes above 1.1 Vp at 15/18/20 km: 14/11/2. Median intervals: 3.04, 3.17, 8.64 yr. Durations and slip are tabulated in the report; exact pulse sequence is not recovered.                                                                       |
| F5b   | Slip velocity versus accepted step        | Main Fig. 5; Eqs. (1)-(9); Table 1; $T=1e7$ s, $q_0=3.3e-10$ m/s          | <code>python3 scripts/panel.py F5b --data</code> | <code>python3 scripts/plot.py F5b</code> | partial | Complete aseismic episodes above 1.1 Vp at 15/18/20 km: 14/11/2. Median intervals: 3.04, 3.17, 8.64 yr. Durations and slip are tabulated in the report; exact pulse sequence is not recovered.                                                                       |
| F5c   | Effective normal stress                   | Main Fig. 5; Eqs. (1)-(9); Table 1; $T=1e7$ s, $q_0=3.3e-10$ m/s          | <code>python3 scripts/panel.py F5c --data</code> | <code>python3 scripts/plot.py F5c</code> | partial | Complete aseismic episodes above 1.1 Vp at 15/18/20 km: 14/11/2. Median intervals: 3.04, 3.17, 8.64 yr. Durations and slip are tabulated in the report; exact pulse sequence is not recovered.                                                                       |
| F5d   | Permeability                              | Main Fig. 5; Eqs. (1)-(9); Table 1; $T=1e7$ s, $q_0=3.3e-10$ m/s          | <code>python3 scripts/panel.py F5d --data</code> | <code>python3 scripts/plot.py F5d</code> | partial | Complete aseismic episodes above 1.1 Vp at 15/18/20 km: 14/11/2. Median intervals: 3.04, 3.17, 8.64 yr. Durations and slip are tabulated in the report; exact pulse sequence is not recovered.                                                                       |
| F6    | Upward migration of the $ V =V_p$ contour | Main Fig. 6 and Discussion; reported rates 2.50, 0.38, 0.12, 0.03 km/year | <code>python3 scripts/panel.py F6 --data</code>  | <code>python3 scripts/plot.py F6</code>  | partial | Leading/deepest contour medians at 13–20 km, short through verylong: 1.84e+03/1.53e+03, 647/967, 187/242, 168/168 m/year; published targets 2500, 380, 120, 30. Both contour choices are retained. Baseline shallow medians: 7.27e+03/unmeasured m/year versus 4560. |
| S1a   | Slip velocity at 5, 10, 15, and 20 km     | Supplementary Fig. 1; main Eqs. (1)-(9)                                   | <code>python3 scripts/panel.py S1a --data</code> | <code>python3 scripts/plot.py S1a</code> | partial | Exact friction and initialization are not fully specified; numerical trajectory differs.                                                                                                                                                                             |
| S1b   | Permeability at 5, 10, 15, and 20 km      | Supplementary Fig. 1; main Eqs. (1)-(9)                                   | <code>python3 scripts/panel.py S1b --data</code> | <code>python3 scripts/plot.py S1b</code> | partial | Exact friction and initialization are not fully specified; numerical trajectory differs.                                                                                                                                                                             |
| S1c   | Upward flux at 5, 10, 15, and 20 km       | Supplementary Fig. 1; main Eqs. (1)-(9)                                   | <code>python3 scripts/panel.py S1c --data</code> | <code>python3 scripts/plot.py S1c</code> | partial | Exact friction and initialization are not fully specified; numerical trajectory differs.                                                                                                                                                                             |

| Panel | Quantity                                                      | Documentation                                                                        | Data command                                     | Plot command                             | Status  | Discrepancy                                                                                                                                                                    |
|-------|---------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------|------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S1d   | Effective stress at 5, 10, 15, and 20 km                      | Supplementary Fig. 1; main Eqs. (1)-(9)                                              | <code>python3 scripts/panel.py S1d --data</code> | <code>python3 scripts/plot.py S1d</code> | partial | Exact friction and initialization are not fully specified; numerical trajectory differs.                                                                                       |
| S2a   | Alternative cycle: slip velocity in time and step coordinates | Visible panels a,b of Supplementary Fig. 2; caption additionally mentions absent c,d | <code>python3 scripts/panel.py S2a --data</code> | <code>python3 scripts/plot.py S2a</code> | partial | Exact friction and initialization are not fully specified; numerical trajectory differs.                                                                                       |
| S2b   | Alternative cycle: slip velocity in time and step coordinates | Visible panels a,b of Supplementary Fig. 2; caption additionally mentions absent c,d | <code>python3 scripts/panel.py S2b --data</code> | <code>python3 scripts/plot.py S2b</code> | partial | Exact friction and initialization are not fully specified; numerical trajectory differs.                                                                                       |
| S3a   | Slip velocity in time and step coordinates                    | Supplementary Fig. 3; main Eqs. (1)-(9), Table 1                                     | <code>python3 scripts/panel.py S3a --data</code> | <code>python3 scripts/plot.py S3a</code> | partial | Last complete interval 88.4 yr versus approximately 160 yr in Supplementary Fig. 3a (-45%). All-interval median 88.4 yr. Initial state and friction profiles remain uncertain. |
| S3b   | Cumulative slip profiles                                      | Supplementary Fig. 3; main Eqs. (1)-(9), Table 1                                     | <code>python3 scripts/panel.py S3b --data</code> | <code>python3 scripts/plot.py S3b</code> | partial | Last complete interval 88.4 yr versus approximately 160 yr in Supplementary Fig. 3a (-45%). All-interval median 88.4 yr. Initial state and friction profiles remain uncertain. |
| S3c   | Slip velocity in time and step coordinates                    | Supplementary Fig. 3; main Eqs. (1)-(9), Table 1                                     | <code>python3 scripts/panel.py S3c --data</code> | <code>python3 scripts/plot.py S3c</code> | partial | Last complete interval 70.5 yr versus approximately 65 yr in Supplementary Fig. 3c (+8%). All-interval median 63.6 yr. Initial state and friction profiles remain uncertain.   |
| S3d   | Cumulative slip profiles                                      | Supplementary Fig. 3; main Eqs. (1)-(9), Table 1                                     | <code>python3 scripts/panel.py S3d --data</code> | <code>python3 scripts/plot.py S3d</code> | partial | Last complete interval 70.5 yr versus approximately 65 yr in Supplementary Fig. 3c (+8%). All-interval median 63.6 yr. Initial state and friction profiles remain uncertain.   |
| S4a   | Effective normal stress history                               | Supplementary Fig. 4; main Eqs. (1)-(4), Table 1                                     | <code>python3 scripts/panel.py S4a --data</code> | <code>python3 scripts/plot.py S4a</code> | partial | At 10 km, stress range 16.5 MPa and permeability ratio 9.64. The target is reduced valving with long healing; event sequence and mesh convergence remain uncertain.            |
| S4b   | Effective normal stress depth profiles                        | Supplementary Fig. 4; main Eqs. (1)-(4), Table 1                                     | <code>python3 scripts/panel.py S4b --data</code> | <code>python3 scripts/plot.py S4b</code> | partial | At 10 km, stress range 16.5 MPa and permeability ratio 9.64. The target is reduced valving with long healing; event sequence and mesh convergence remain uncertain.            |
| S4c   | Permeability history                                          | Supplementary Fig. 4; main Eqs. (1)-(4), Table 1                                     | <code>python3 scripts/panel.py S4c --data</code> | <code>python3 scripts/plot.py S4c</code> | partial | At 10 km, stress range 16.5 MPa and permeability ratio 9.64. The target is reduced valving with long healing; event sequence and mesh convergence remain uncertain.            |
| S4d   | Permeability depth profiles                                   | Supplementary Fig. 4; main Eqs. (1)-(4), Table 1                                     | <code>python3 scripts/panel.py S4d --data</code> | <code>python3 scripts/plot.py S4d</code> | partial | At 10 km, stress range 16.5 MPa and permeability ratio 9.64. The target is reduced valving with long healing; event sequence and mesh convergence remain uncertain.            |

| Panel | Quantity                                   | Documentation                                    | Data command                                     | Plot command                             | Status  | Discrepancy                                                                                                                                                                    |
|-------|--------------------------------------------|--------------------------------------------------|--------------------------------------------------|------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S4e   | Upward fluid flux history                  | Supplementary Fig. 4; main Eqs. (1)-(4), Table 1 | <code>python3 scripts/panel.py S4e --data</code> | <code>python3 scripts/plot.py S4e</code> | partial | At 10 km, stress range 16.5 MPa and permeability ratio 9.64. The target is reduced valving with long healing; event sequence and mesh convergence remain uncertain.            |
| S4f   | Upward fluid flux depth profiles           | Supplementary Fig. 4; main Eqs. (1)-(4), Table 1 | <code>python3 scripts/panel.py S4f --data</code> | <code>python3 scripts/plot.py S4f</code> | partial | At 10 km, stress range 16.5 MPa and permeability ratio 9.64. The target is reduced valving with long healing; event sequence and mesh convergence remain uncertain.            |
| S5a   | Slip velocity in time and step coordinates | Supplementary Fig. 5; main Eqs. (1)-(9), Table 1 | <code>python3 scripts/panel.py S5a --data</code> | <code>python3 scripts/plot.py S5a</code> | partial | Last complete interval 95.1 yr versus approximately 160 yr in Supplementary Fig. 5a (-41%). All-interval median 95.2 yr. Initial state and friction profiles remain uncertain. |
| S5b   | Cumulative slip profiles                   | Supplementary Fig. 5; main Eqs. (1)-(9), Table 1 | <code>python3 scripts/panel.py S5b --data</code> | <code>python3 scripts/plot.py S5b</code> | partial | Last complete interval 95.1 yr versus approximately 160 yr in Supplementary Fig. 5a (-41%). All-interval median 95.2 yr. Initial state and friction profiles remain uncertain. |
| S5c   | Slip velocity in time and step coordinates | Supplementary Fig. 5; main Eqs. (1)-(9), Table 1 | <code>python3 scripts/panel.py S5c --data</code> | <code>python3 scripts/plot.py S5c</code> | partial | Last complete interval 174 yr versus approximately 150 yr in Supplementary Fig. 5c (+16%). All-interval median 179 yr. Initial state and friction profiles remain uncertain.   |
| S5d   | Cumulative slip profiles                   | Supplementary Fig. 5; main Eqs. (1)-(9), Table 1 | <code>python3 scripts/panel.py S5d --data</code> | <code>python3 scripts/plot.py S5d</code> | partial | Last complete interval 174 yr versus approximately 150 yr in Supplementary Fig. 5c (+16%). All-interval median 179 yr. Initial state and friction profiles remain uncertain.   |
| S6a   | Effective normal stress history            | Supplementary Fig. 6; main Eqs. (1)-(4), Table 1 | <code>python3 scripts/panel.py S6a --data</code> | <code>python3 scripts/plot.py S6a</code> | partial | At 10 km, stress range 6.82 MPa and permeability ratio 1.77. The target is reduced valving with long healing; event sequence and mesh convergence remain uncertain.            |
| S6b   | Effective normal stress depth profiles     | Supplementary Fig. 6; main Eqs. (1)-(4), Table 1 | <code>python3 scripts/panel.py S6b --data</code> | <code>python3 scripts/plot.py S6b</code> | partial | At 10 km, stress range 6.82 MPa and permeability ratio 1.77. The target is reduced valving with long healing; event sequence and mesh convergence remain uncertain.            |
| S6c   | Permeability history                       | Supplementary Fig. 6; main Eqs. (1)-(4), Table 1 | <code>python3 scripts/panel.py S6c --data</code> | <code>python3 scripts/plot.py S6c</code> | partial | At 10 km, stress range 6.82 MPa and permeability ratio 1.77. The target is reduced valving with long healing; event sequence and mesh convergence remain uncertain.            |
| S6d   | Permeability depth profiles                | Supplementary Fig. 6; main Eqs. (1)-(4), Table 1 | <code>python3 scripts/panel.py S6d --data</code> | <code>python3 scripts/plot.py S6d</code> | partial | At 10 km, stress range 6.82 MPa and permeability ratio 1.77. The target is reduced valving with long healing; event sequence and mesh convergence remain uncertain.            |

| Panel | Quantity                         | Documentation                                    | Data command                                     | Plot command                             | Status  | Discrepancy                                                                                                                                                         |
|-------|----------------------------------|--------------------------------------------------|--------------------------------------------------|------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S6e   | Upward fluid flux history        | Supplementary Fig. 6; main Eqs. (1)-(4), Table 1 | <code>python3 scripts/panel.py S6e --data</code> | <code>python3 scripts/plot.py S6e</code> | partial | At 10 km, stress range 6.82 MPa and permeability ratio 1.77. The target is reduced valving with long healing; event sequence and mesh convergence remain uncertain. |
| S6f   | Upward fluid flux depth profiles | Supplementary Fig. 6; main Eqs. (1)-(4), Table 1 | <code>python3 scripts/panel.py S6f --data</code> | <code>python3 scripts/plot.py S6f</code> | partial | At 10 km, stress range 6.82 MPa and permeability ratio 1.77. The target is reduced valving with long healing; event sequence and mesh convergence remain uncertain. |

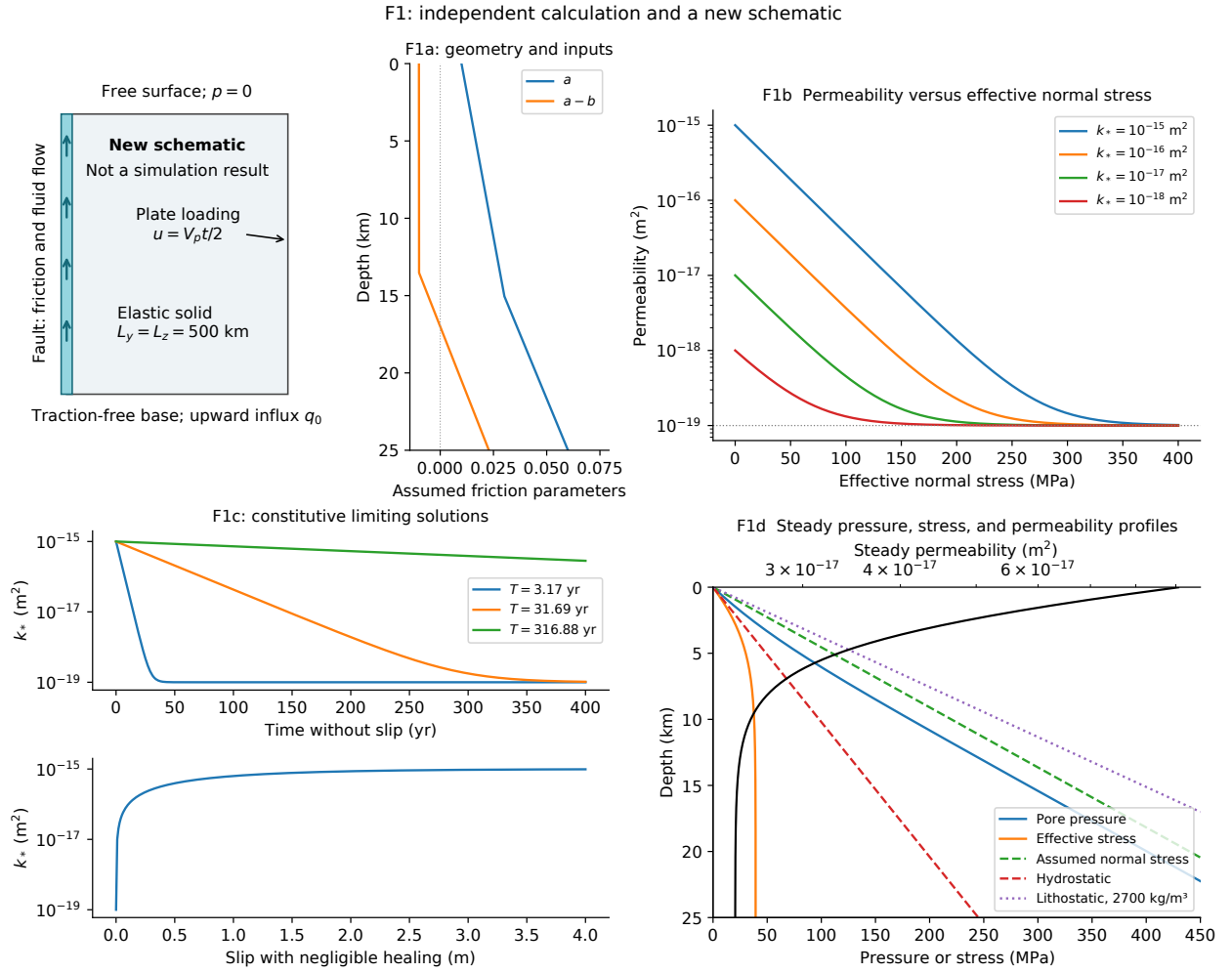


Figure 1: Independent outputs corresponding to F1. F1a: Geometry schematic and assumed friction profiles; F1b: Permeability versus effective normal stress; F1c: Healing and slip-enhancement limiting solutions; F1d: Steady pressure, stress, and permeability profiles. Units and color scales are shown on the panels. Parameter files and selection rules are specified in the provenance sidecars. Panel F1a is a new schematic, not a simulated reproduction. Comparison target: Zhu et al. [1].

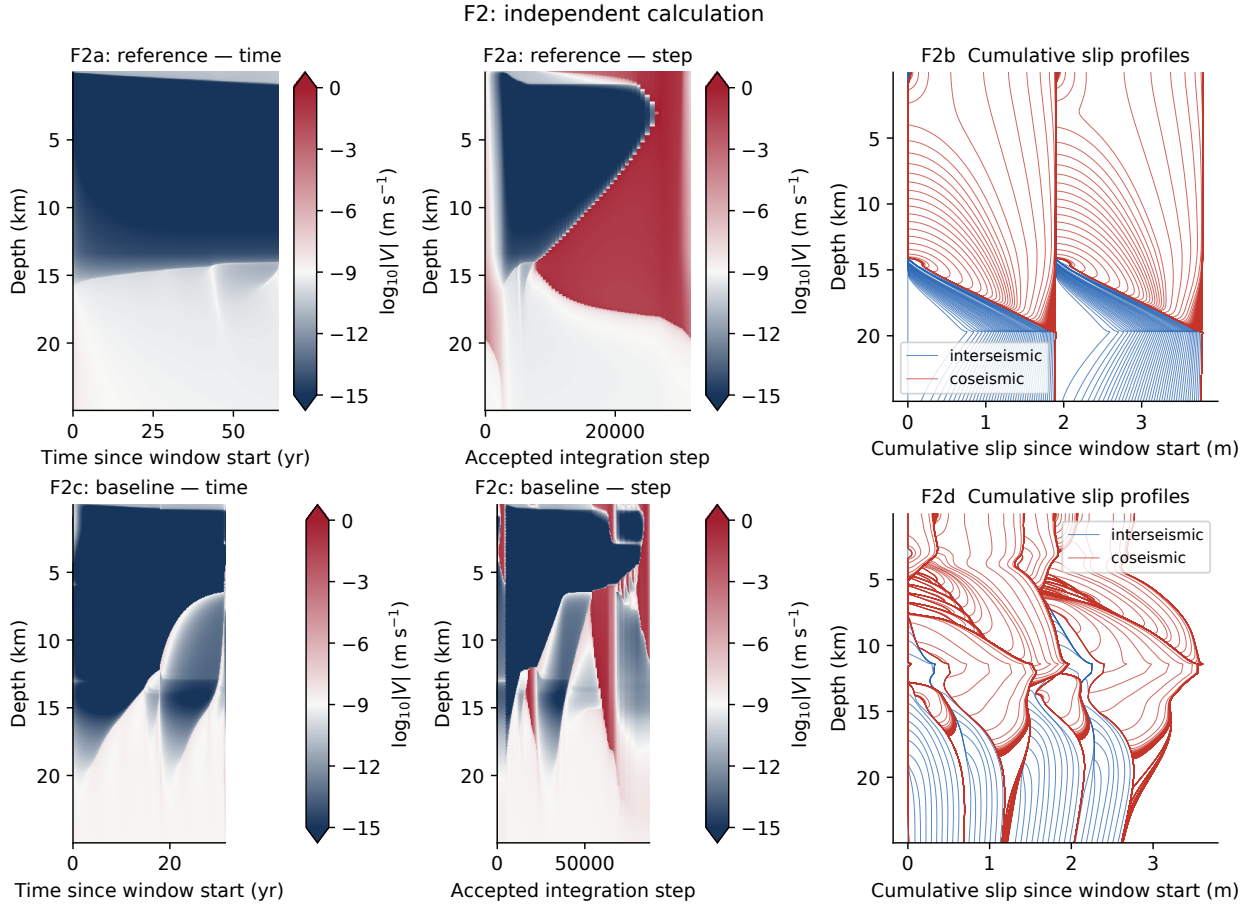


Figure 2: Independent outputs corresponding to F2. F2a: Slip velocity in time and step coordinates; F2b: Cumulative slip profiles; F2c: Slip velocity in time and step coordinates; F2d: Cumulative slip profiles. Units and color scales are shown on the panels. Parameter files and selection rules are specified in the provenance sidecars. Cases: baseline ( $T=1\text{e}+08$  s; coupled pressure), reference ( $T=1\text{e}+08$  s; fixed pressure). These panels are partial comparisons. Comparison target: Zhu et al. [1].

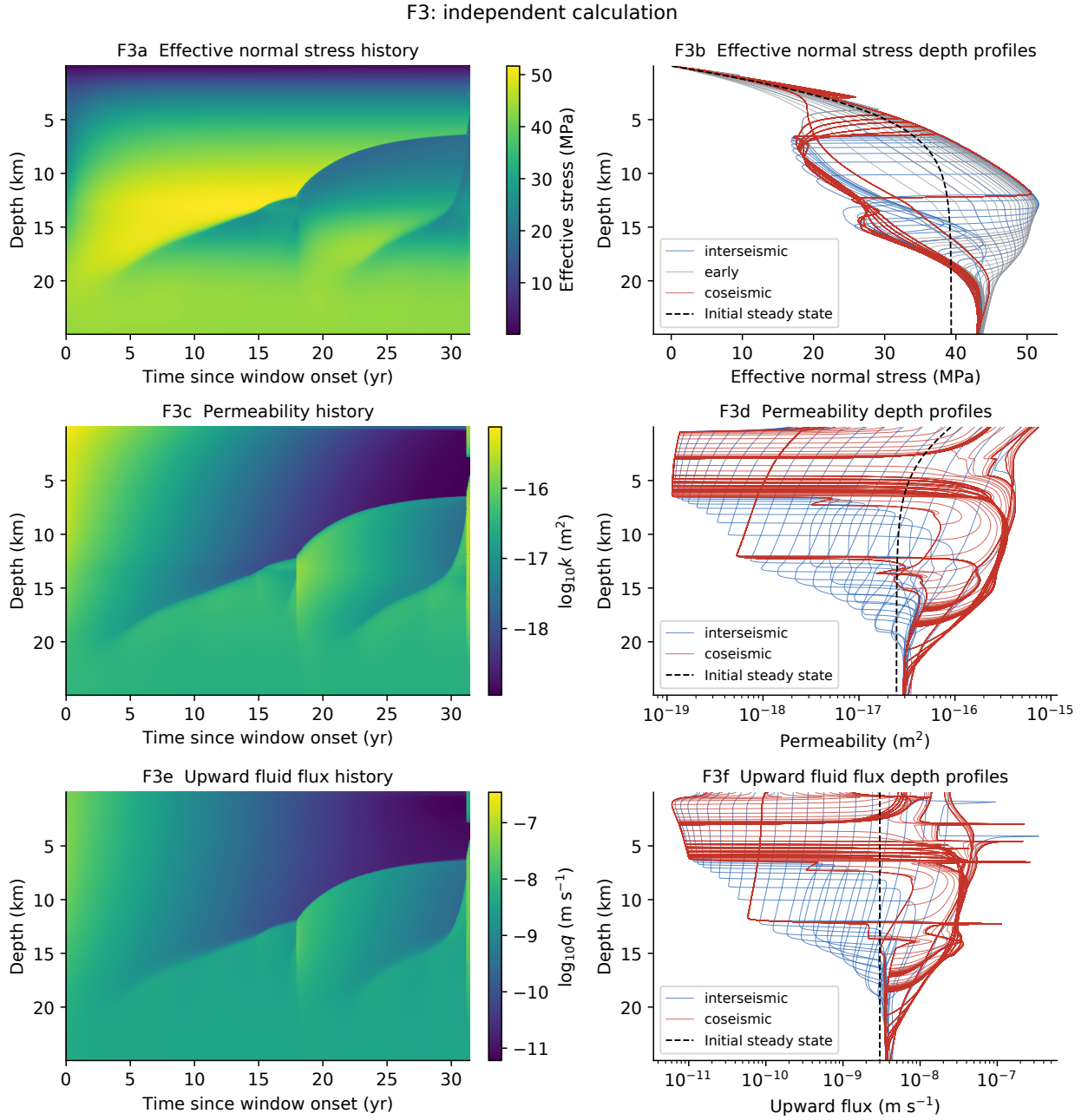


Figure 3: Independent outputs corresponding to F3. F3a: Effective normal stress history; F3b: Effective normal stress depth profiles; F3c: Permeability history; F3d: Permeability depth profiles; F3e: Upward fluid flux history; F3f: Upward fluid flux depth profiles. Units and color scales are shown on the panels. Parameter files and selection rules are specified in the provenance sidecars. Cases: baseline ( $T=1e+08$  s; coupled pressure). These panels are partial comparisons. Comparison target: Zhu et al. [1].

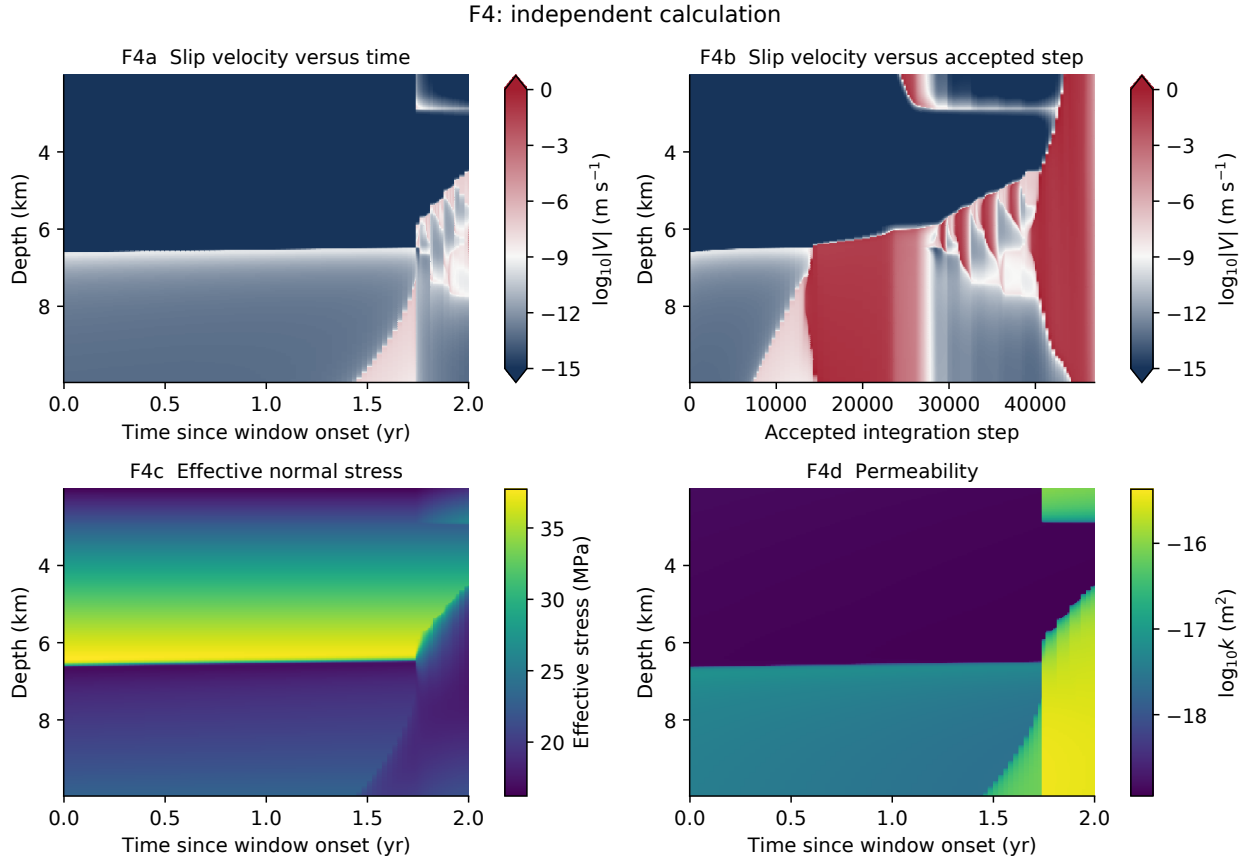


Figure 4: Independent outputs corresponding to F4. F4a: Slip velocity versus time; F4b: Slip velocity versus accepted step; F4c: Effective normal stress; F4d: Permeability. Units and color scales are shown on the panels. Parameter files and selection rules are specified in the provenance sidecars. Cases: baseline ( $T=1e+08$  s; coupled pressure). These panels are partial comparisons. Comparison target: Zhu et al. [1].

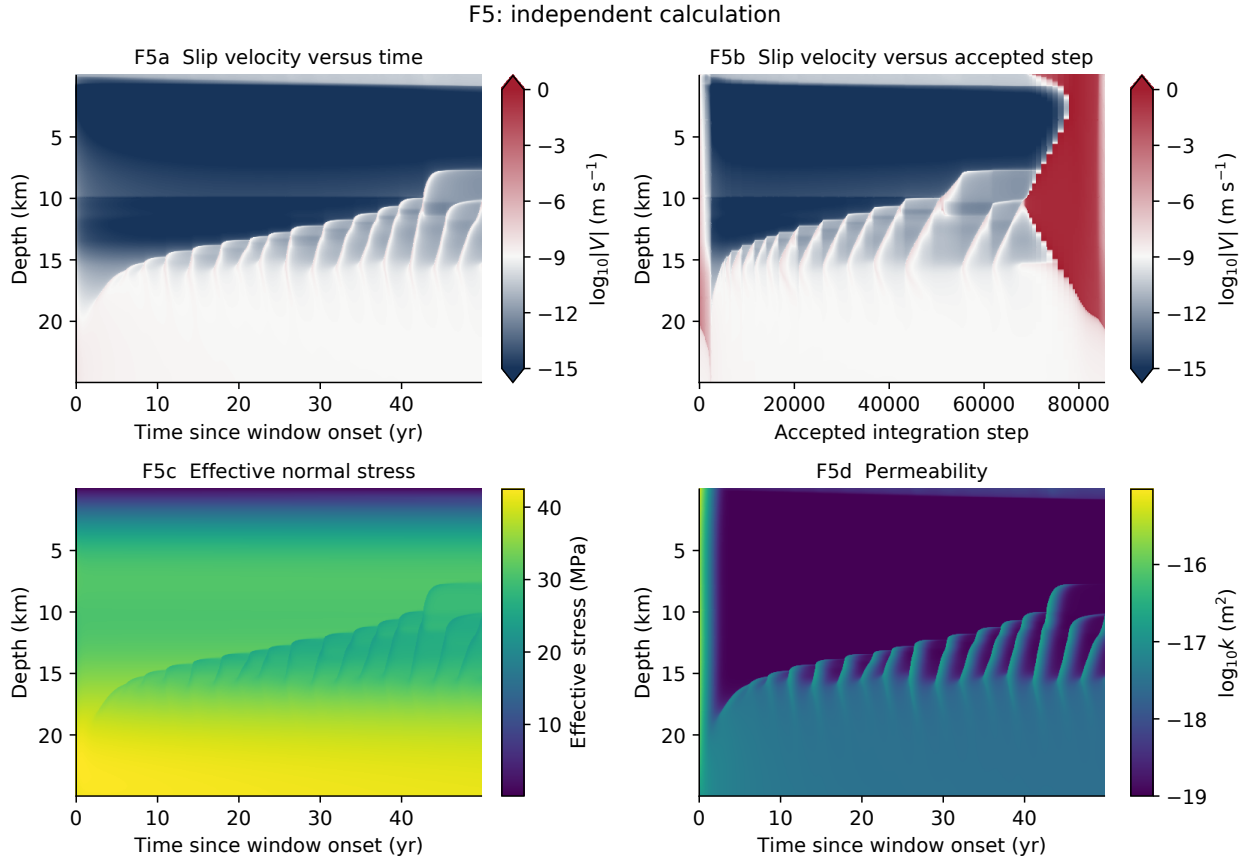


Figure 5: Independent outputs corresponding to F5. F5a: Slip velocity versus time; F5b: Slip velocity versus accepted step; F5c: Effective normal stress; F5d: Permeability. Units and color scales are shown on the panels. Parameter files and selection rules are specified in the provenance sidecars. Cases: short ( $T=1e+07$  s; coupled pressure). These panels are partial comparisons. Comparison target: Zhu et al. [1].

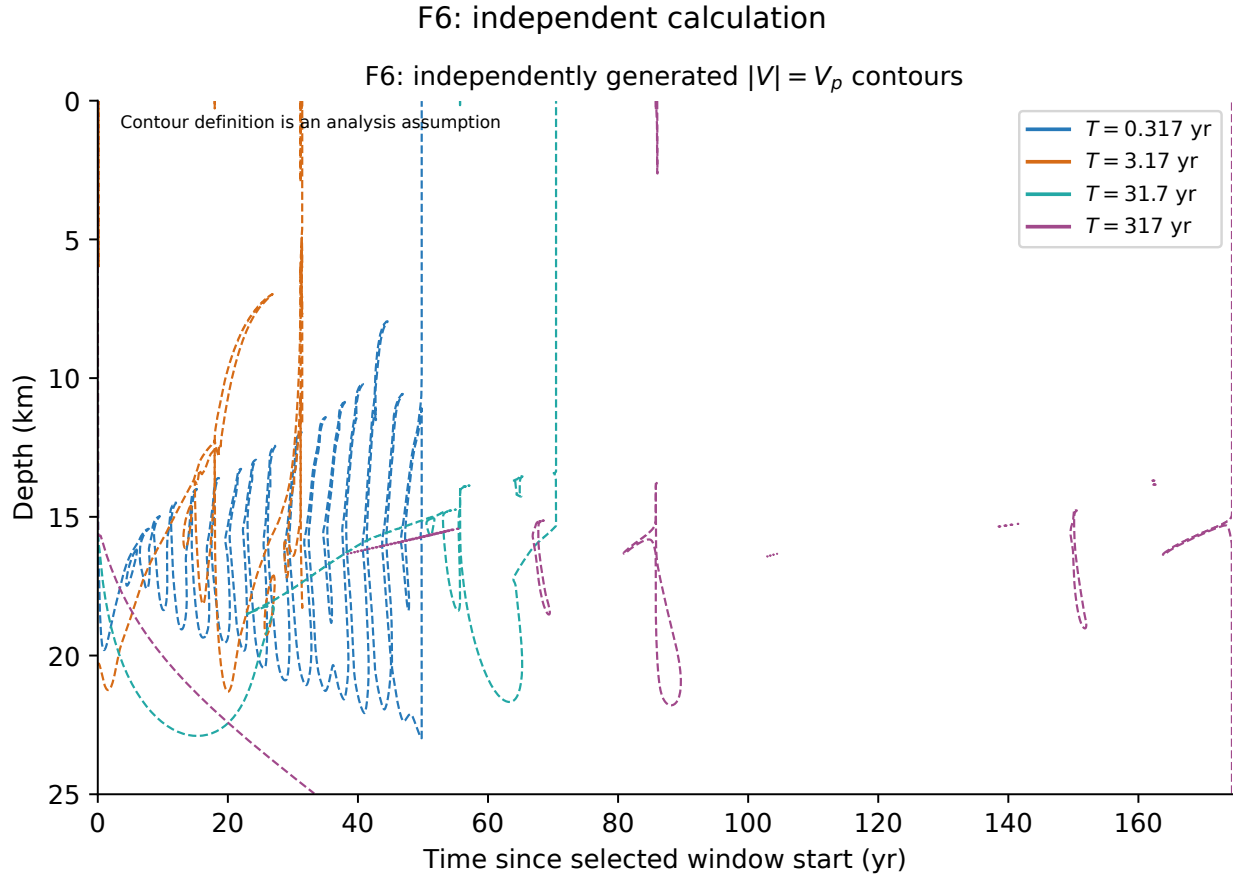


Figure 6: Independent outputs corresponding to F6. F6: Upward migration of the  $|V|=V_p$  contour. Units and color scales are shown on the panels. Parameter files and selection rules are specified in the provenance sidecars. Cases: baseline ( $T=1e+08$  s; coupled pressure), long ( $T=1e+09$  s; coupled pressure), short ( $T=1e+07$  s; coupled pressure), verylong ( $T=1e+10$  s; coupled pressure). These panels are partial comparisons. Comparison target: Zhu et al. [1].

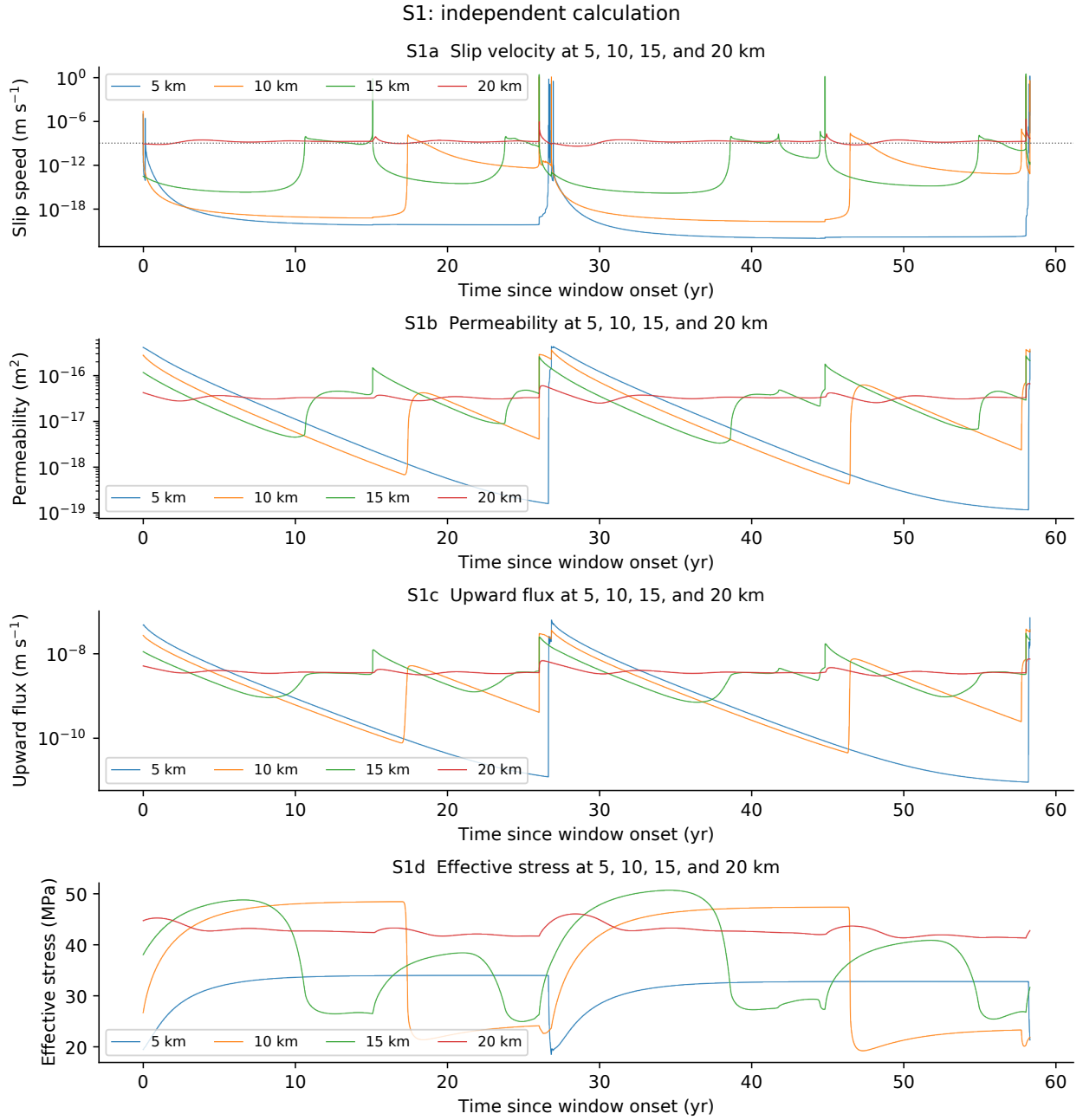


Figure 7: Independent outputs corresponding to S1. S1a: Slip velocity at 5, 10, 15, and 20 km; S1b: Permeability at 5, 10, 15, and 20 km; S1c: Upward flux at 5, 10, 15, and 20 km; S1d: Effective stress at 5, 10, 15, and 20 km. Units and color scales are shown on the panels. Parameter files and selection rules are specified in the provenance sidecars. Cases: baseline ( $T=1e+08$  s; coupled pressure). These panels are partial comparisons. Comparison target: Zhu et al. [1].

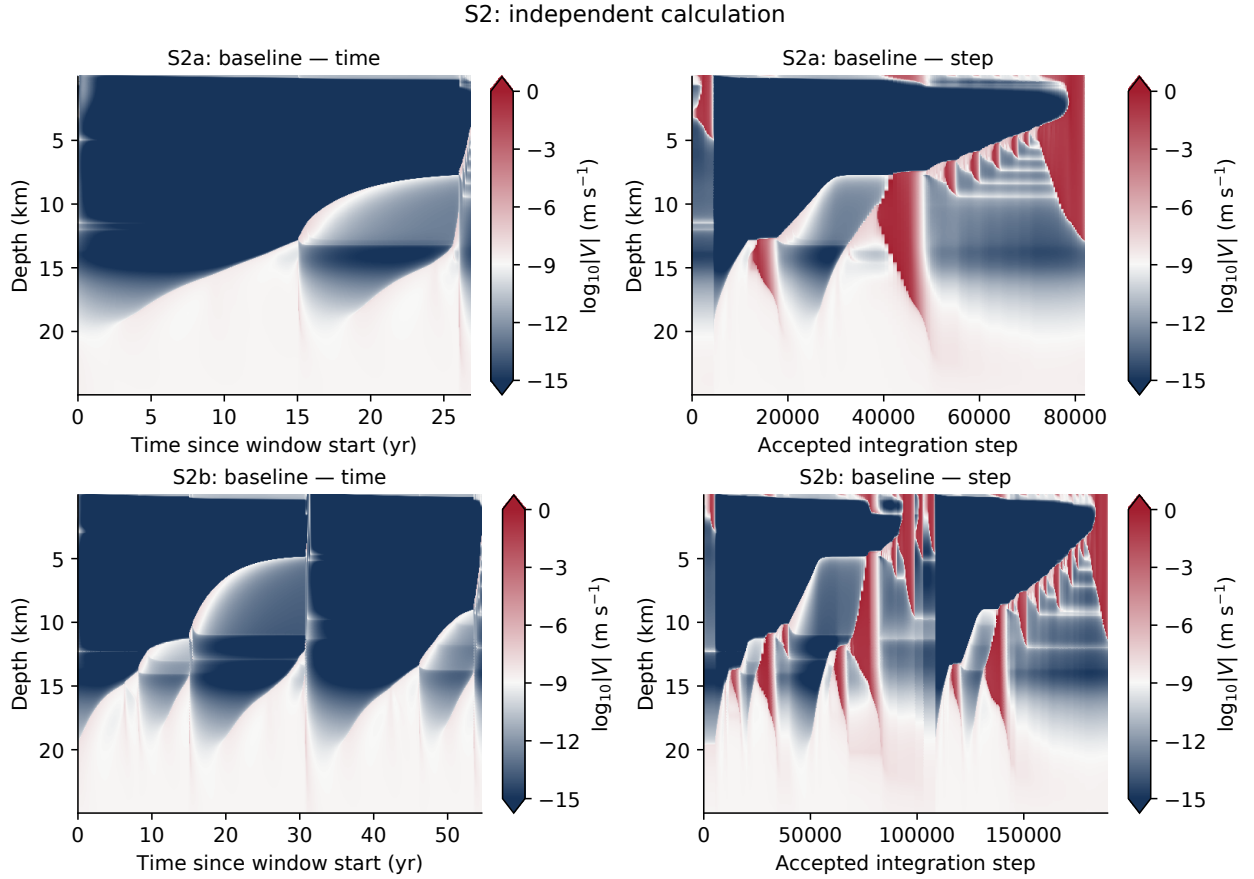


Figure 8: Independent outputs corresponding to S2. S2a: Alternative cycle: slip velocity in time and step coordinates; S2b: Alternative cycle: slip velocity in time and step coordinates. Units and color scales are shown on the panels. Parameter files and selection rules are specified in the provenance sidecars. Cases: baseline ( $T=1\text{e}+08$  s; coupled pressure). These panels are partial comparisons. Comparison target: Zhu et al. [1].

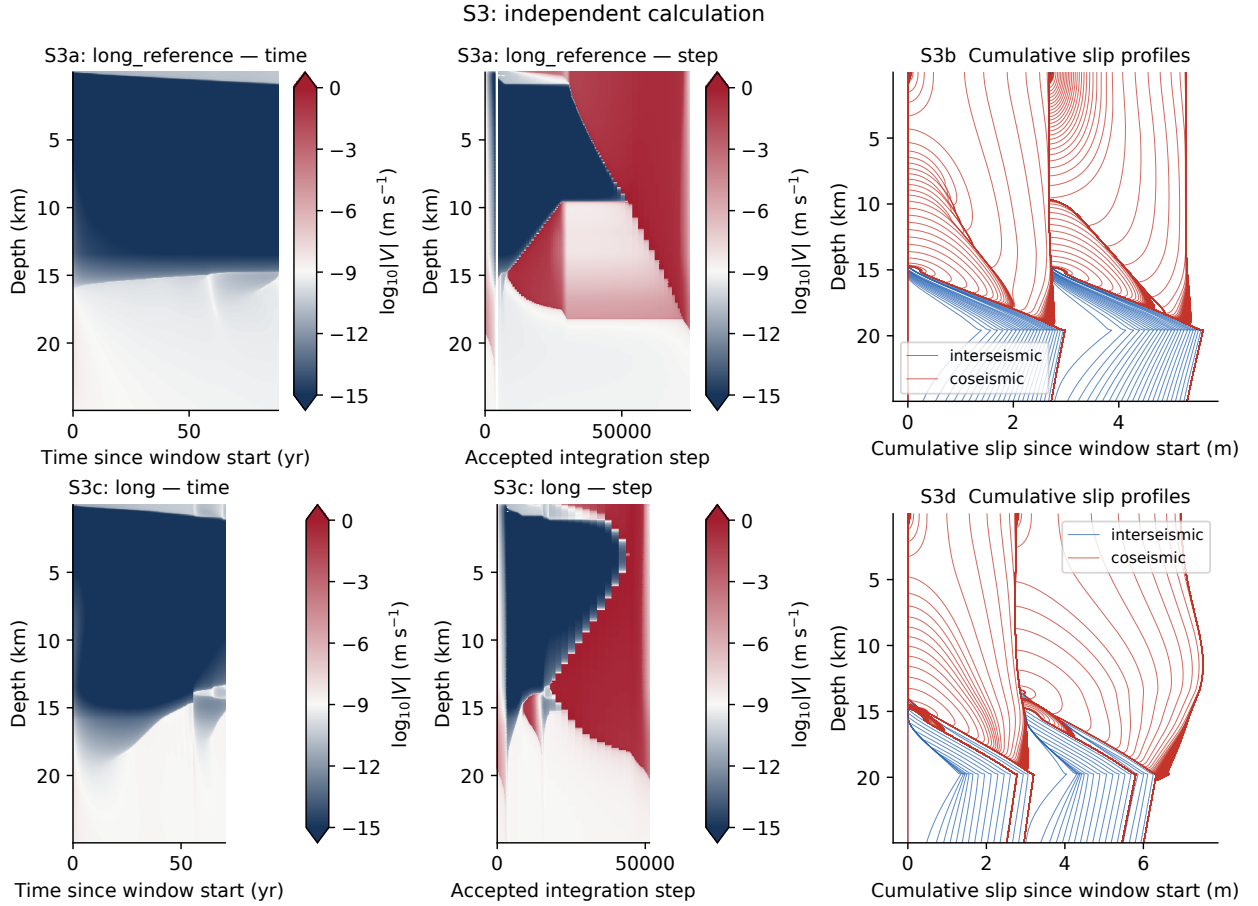


Figure 9: Independent outputs corresponding to S3. S3a: Slip velocity in time and step coordinates; S3b: Cumulative slip profiles; S3c: Slip velocity in time and step coordinates; S3d: Cumulative slip profiles. Units and color scales are shown on the panels. Parameter files and selection rules are specified in the provenance sidecars. Cases: long ( $T=1e+09$  s; coupled pressure), long\_reference ( $T=1e+09$  s; fixed pressure). These panels are partial comparisons. Comparison target: Zhu et al. [1].

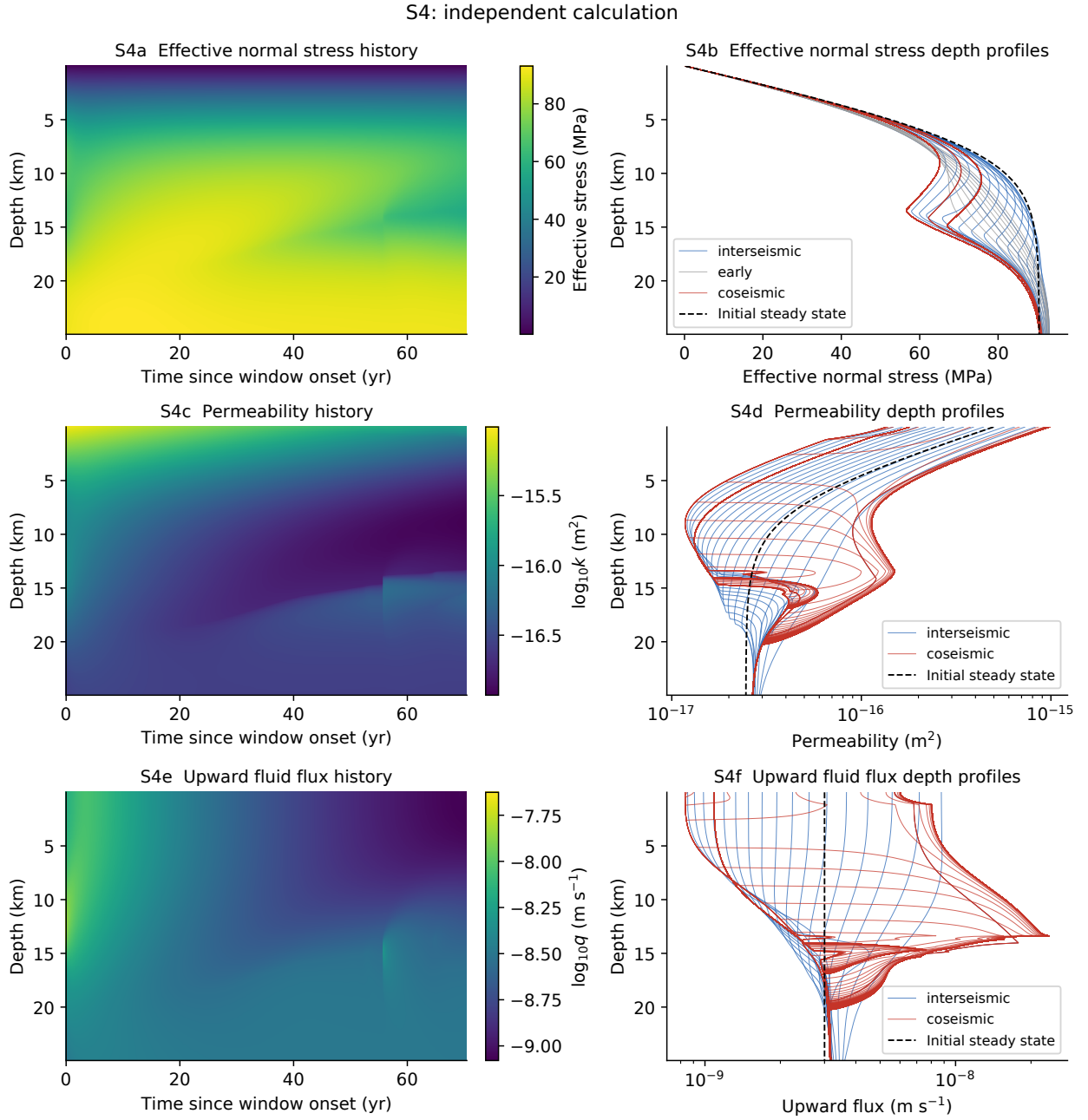


Figure 10: Independent outputs corresponding to S4. S4a: Effective normal stress history; S4b: Effective normal stress depth profiles; S4c: Permeability history; S4d: Permeability depth profiles; S4e: Upward fluid flux history; S4f: Upward fluid flux depth profiles. Units and color scales are shown on the panels. Parameter files and selection rules are specified in the provenance sidecars. Cases: long ( $T=1\text{e}+09$  s; coupled pressure). These panels are partial comparisons. Comparison target: Zhu et al. [1].

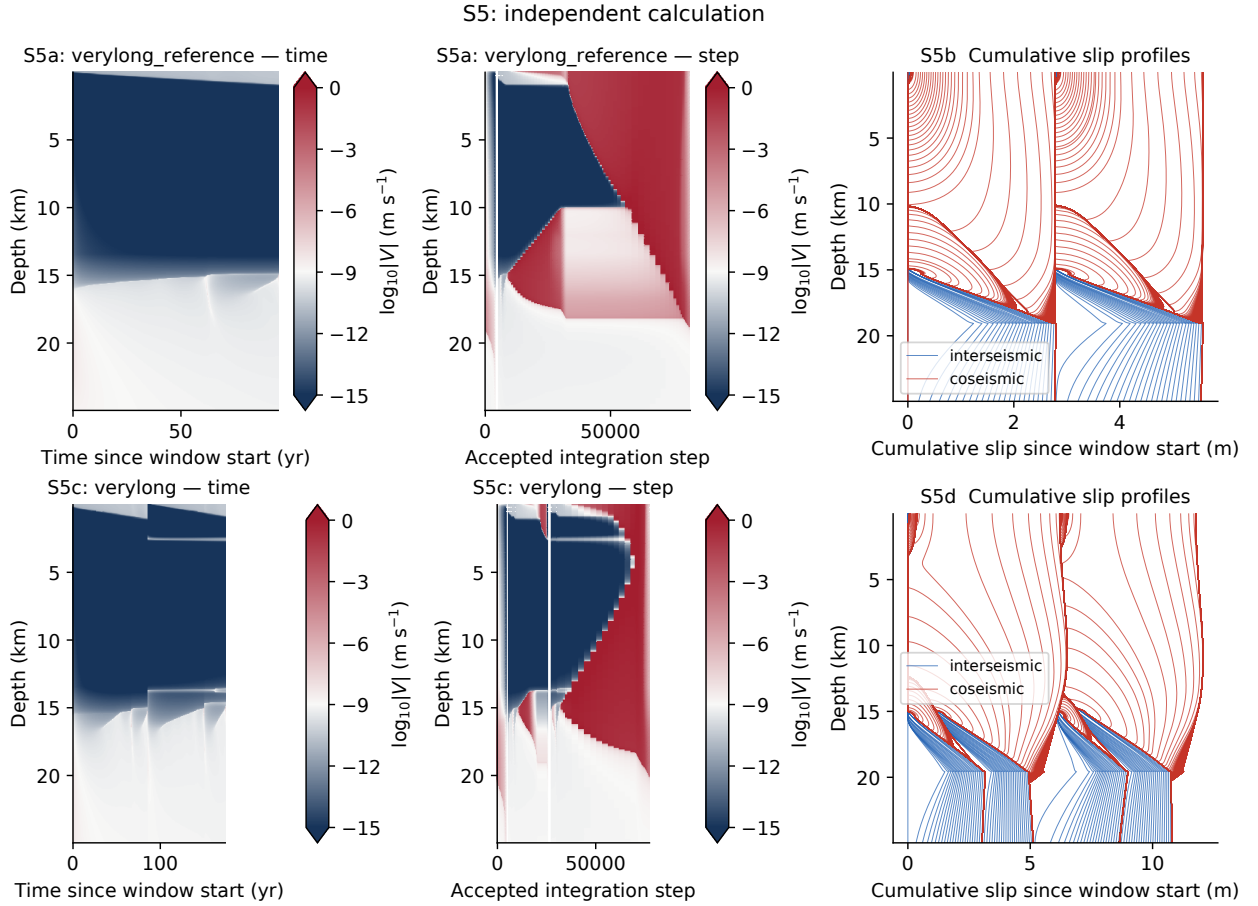


Figure 11: Independent outputs corresponding to S5. S5a: Slip velocity in time and step coordinates; S5b: Cumulative slip profiles; S5c: Slip velocity in time and step coordinates; S5d: Cumulative slip profiles. Units and color scales are shown on the panels. Parameter files and selection rules are specified in the provenance sidecars. Cases: verylong ( $T=1e+10$  s; coupled pressure), verylong\_reference ( $T=1e+10$  s; fixed pressure). These panels are partial comparisons. Comparison target: Zhu et al. [1].

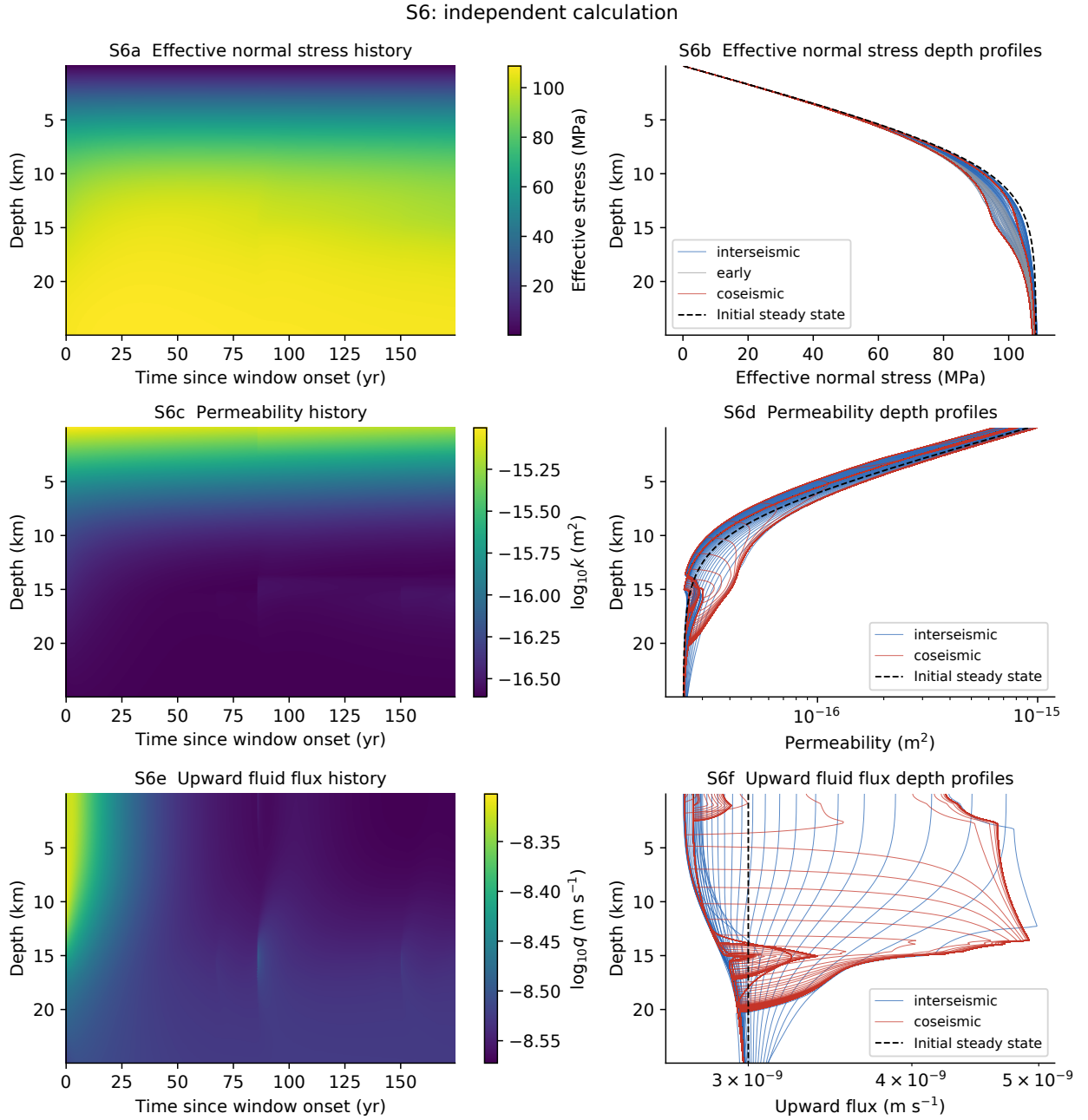


Figure 12: Independent outputs corresponding to S6. S6a: Effective normal stress history; S6b: Effective normal stress depth profiles; S6c: Permeability history; S6d: Permeability depth profiles; S6e: Upward fluid flux history; S6f: Upward fluid flux depth profiles. Units and color scales are shown on the panels. Parameter files and selection rules are specified in the provenance sidecars. Cases: verylong ( $T=1e+10$  s; coupled pressure). These panels are partial comparisons. Comparison target: Zhu et al. [1].

## 6 Reproduction and remaining work

The full project command builds and tests the C++ solver, runs all twelve cases, extracts every panel, compiles the final report, and verifies its source bundle:

```
python3 -m pip install -r requirements.txt
python3 scripts/reproduce.py
```

That full-duration run is ongoing at this snapshot. This interim report is regenerated from completed, audited case metadata and generated panel packages:

```
python3 scripts/interim_report.py
```

The command compiles `report/interim.pdf`, packages its standalone LaTeX source and all included figures, and verifies compilation of the extracted bundle. It records commands, exit codes, input hashes, and output hashes in `data/interim_report_verification.json`. The source archive is `report/interim-latex-source.zip`; the unexpanded source is already a single file, `report/interim.tex`. All scientific arrays and their independent generation commands are available in the repository.

Completion still requires the remaining calculations, any pending panels listed in the inventory, common-duration validation comparisons, the full scientific report, visual and numerical artifact review, execution of the final regeneration procedure, and publication of those final deliverables. This snapshot makes the completed scientific results reviewable without declaring that work finished.

## References

- [1] Zhu, W., K. L. Allison, E. M. Dunham, and Y. Yang (2020). Fault valving and pore pressure evolution in simulations of earthquake sequences and aseismic slip. *Nature Communications*, 11, 4833. doi:10.1038/s41467-020-18598-z. Includes the published Supplementary Information.
- [2] Allison, K. L., and E. M. Dunham (2018). Earthquake cycle simulations with rate-and-state friction and power-law viscoelasticity. *Tectonophysics*, 733, 232–256. doi:10.1016/j.tecto.2017.10.021.
- [3] Kennedy, C. A., and M. H. Carpenter (2001). Additive Runge–Kutta schemes for convection–diffusion–reaction equations. NASA/TM-2001-211038, Appendix C. NASA technical report.